

Soccer Data Analytics Hackathon



supported by  PySport

📅 February 27–28, 2026

📍 Northeastern University / Network Science Institute

🌐 <https://sites.google.com/view/neusportsanalytics>

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Overview. This handout contains the challenge prompts, deliverables, timeline, judging rubric, and logistics for the *Soccer Data Analytics Hackathon*. You may choose **one** of the two prompts below (Starting Eleven or Player Valuation). Teams of 2–5 students are recommended.

1 Data & Tools

- **Primary dataset:** IMPECT Open Data events, lineups, players, teams for 306 German Bundesliga matches from the 2023/24 season. Non-commercial license applies (see [License](#)).
- **Suggested tooling:** Python ([kloppy](#), [polars](#), [pandas](#), [numpy](#), [mplsoccer](#), [databallpy](#), [networkx](#)), or R ([tidyverse](#)/[igraph](#)). Any language is fine; reproducibility is required.
- **Reproducibility:** Put code on GitHub with a clear README.md and environment file (`environment.yml` or `requirements.txt`), and include an ([open-source license](#) (e.g., [MIT](#) or [Apache 2.0](#)))
 - You can fork the template repo available [here](#).

2 Choose One Prompt

2.1 (A) Starting Eleven Lineup Construction

Goal: Recommend a starting eleven (and/or substitution plan) to maximize team cohesion and ball progression for a chosen team/competition.

Example Approach: Build a directed, weighted player-to-player pass network from event data (edges represent passes, weights incorporate completion and progression). Combine with roles/positions from lineups to score any lineup's fit under a formation constraint.

- Analyze network structure.
- Compare alternative starting elevens, justify trade-offs and role compatibility.
- Visualize pass maps and network overlays, keep figures readable.

2.2 (B) Build a Transparent Player Valuation Metric

Goal: Define an interpretable attacking or defensive metric using events, compare players within one competition/season.

Examples:

- Attacking: progressive passes into the box, pre-shot value, etc.
- Defending: pressures/tackles/interceptions per opponent possession, share of final-third passes, etc.

Report the definition, a leaderboard, and one case study. Discuss limitations.

3 Deliverables (Due on Friday, February 27, 2026)

1. **Slide deck** (PDF, max 8 slides, clear labels and legends).
2. **GitHub repository** with code, README, and environment.

4 Timeline

Release & Data Primer	Monday, November 3, 2025
Registration Deadline	Wednesday, December 31, 2025
Checkpoint (draft slide deck and/or repo link)	Monday, February 2, 2026
Final Work Session & Judging	Friday, February 27, 2026
Industry Talks & Awards	Saturday, February 28, 2026

5 Judging Rubric (100 points)

Category	Pts	What we look for
Problem framing & soccer context	10	Clear question, stakeholder value, appropriate scope.
Data engineering & correctness	15	Proper joins (events/matches/lineups), handling subs/missingness, reproducible pipeline.
Methodology quality	15	Sound metric/model design, reasonable assumptions.
Validation & robustness	15	Testing across matches/players, uncertainty assessment, if feasible.
Results & insight	15	Actionable takeaways, not just charts, explain disagreements with intuition.
Communication & visualization	15	Readable figures, concise narrative (6–8 min).
Reproducibility & ethics	15	Public repo, license compliance, disclosure of AI assistance ¹ .

6 Logistics & Presentation

- Presentation length: 6–8 minutes + 2 minutes Q&A.
- Submit slides (PDF) and repo link by the posted deadline.
- Name files `TeamName_Hackathon2026.pdf` etc.

¹For example, the template for this rubric was generated by GPT-5 and then edited substantially

7 Getting Started: How to Use the Data with Kloppy

There is a template GitHub repository available [here](#) which houses a Jupyter notebook that can also get you started using Kloppy.

Kloppy by PySport is *the* industry standard open-source football data standardization package. Kloppy simplifies football data processing by offering a single place to **load**, **filter**, **transform** and **export** your football data in a standardized way. To get started with the open dataset simply,

1. Install Kloppy

```
pip install kloppy>=3.18.0
```

2. Load the data

```
from kloppy import impect

events = impect.load_open_data(match_id=122840)
```

To load other, non-open data use `impect.load()` instead.

3. Filter, Transform and Export

```
df = (
    events.transform(
        to_orientation="STATIC_HOME_AWAY"
    ) # Now, the home team always attacks left to right
    .filter(lambda event: event.period.id == 1) # Only keep
        frames from the first half
    .to_df(
        engine="polars"
    ) # Convert to a Polars DataFrame, or use engine="pandas" for
        a Pandas DataFrame
)
```

License & Ethics. IMPECT Open Data is non-commercial; cite sources. Be transparent about limitations. If using AI tools, document where and how they were used.

8 Contact

Questions? Email northeasternsportsanalytics@gmail.com or see the event page: <https://sites.google.com/view/neusportsanalytics>

Thank you for participating in the Soccer Data Analytics Hackathon!